A Common Electrical-Energy Basis for Comparing Lunar Oxygen and Propellant Production Routes

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Code, data, and all figures are reproducible from the open-source model at https://github.com/dubthree/lunar-propellant-energy-model.

Abstract

Lunar in-situ propellant production is widely proposed as the key to affordable cislunar transport, yet the basic engineering question (which oxygen-extraction route costs the least *energy* per kilogram of product) is hard to answer from the literature, because published figures are reported on incompatible bases: hydrogen reduction as electrical energy, carbothermal reduction as thermal energy, and molten regolith electrolysis (MRE) with no published figure at all. Prior work compares subsets (Taylor and Carrier (1993) reviewed ∼20 processes; Leger et al. (2025) modeled several on a common basis), but no published comparison places all five of these routes, including the thermally-reported carbothermal route and the two electrolysis routes that lack any published energy figure, on a single electrical-equivalent basis with paired uncertainty propagation. We present a small, open, uncertainty-quantified model that places five routes (hydrogen reduction, carbothermal reduction, MRE, molten-salt electrolysis, and polar water-ice mining) on one electrical-equivalent kWh/kg O_2 basis under a single explicit system boundary, with Monte-Carlo propagation of literature parameter ranges, including a continuous reactor heat-loss term grounded in the only measured carbothermal datum (NASA CaRD). We check the framework against the one route with a clean published electrical figure (hydrogen reduction, Leger et al. (2025), 24.3 ± 5.8 kWh/kg LOX): our nominal estimate (24.3) matches Leger's central value, with the Monte-Carlo distribution centered somewhat lower (median ∼21), reported as substantial interval overlap between two independent estimates. Using a paired Monte Carlo that shares common parameters across routes, we report dominance probabilities rather than overlapping error bars. The central finding is that **the polar water-ice route is both the most energy-efficient (cheapest in 99% of paired trials) and the only route that yields a complete LOX+LH_2 propellant**. The reason is structural and, to our knowledge, not previously quantified on a common basis: water mining is the only low-temperature route (sublimation at ∼273 K), so it alone escapes the large, continuous reactor heat-loss penalty (∼63–93 kWh-thermal/kg O_2 in the one measured carbothermal demonstration) that burdens every high-temperature thermochemical or electrochemical route. Once that loss is charged, the high-temperature routes (carbothermal ∼21, molten-salt ∼23, hydrogen reduction ∼24, MRE ∼27 kWh/kg O_2) cluster well above the water route (∼14) with wide, overlapping uncertainty; molten regolith electrolysis is the most likely worst. A sensitivity analysis identifies the single highest-value measurement for each route, we translate energy into required surface power and landed fission-plant mass, and we outline two companion analyses on reusing co-located compute waste heat (which is, fittingly, most useful precisely at the low-temperature water route).

1 Introduction

The case for lunar propellant rests on energy: every process that turns regolith or polar ice into liquid oxygen (and ideally liquid hydrogen) is energy-intensive, and surface power is the binding constraint on any near-term architecture. Yet the comparison that should drive route selection has never been made on equal footing. Three incompatibilities recur in the literature:

1. **Thermal vs electrical energy are conflated.** Carbothermal reduction is reported as delivered thermal energy or "g O_2/kWh thermal"; hydrogen reduction is reported as electrical kWh/kg. On the Moon both ultimately draw on the same scarce electrical supply, but they are not interchangeable at face value.

2. **System boundaries differ.** Some figures include excavation, beneficiation, and liquefaction; others count only the reactor.

3. **Two routes have no published figure at all.** Molten regolith electrolysis and molten-salt electrolysis are described qualitatively; their energy cost is asserted, not quantified.

The consequence is that capital allocation, power-plant sizing, and architecture choice rest on a comparison that does not exist. This is a modeling gap, not a hardware gap, which is precisely why it can be closed now, cheaply, and reproducibly. We build that comparison and quantify its uncertainty.

2 Methods

2.1 Functional unit and system boundary

We compute the **electrical-equivalent energy in kWh per kg of O_2 delivered to cryogenic storage**, and additionally report kWh per kg of total propellant for the one route that co-produces hydrogen. The system boundary is a fixed sequence of stages; every route enables a subset, using the same stage sub-models, so all differences between routes arise from parameters rather than inconsistent accounting:

> excavation/acquisition → beneficiation → heating (sensible, plus fusion for melt routes) → reaction (reduction enthalpy or Faradaic electrolysis) → product cleanup → water electrolysis (where the route produces H_2O) → gas compression → liquefaction (LOX always; LH_2 where hydrogen is retained).

2.2 Thermal-to-electrical conversion

To compare a thermally-reported route against electrically-reported ones, all thermal demand is converted to electrical-equivalent through an explicit electric-to-thermal efficiency parameter (resistive heating, nominal 0.90). This single, visible knob is what makes the comparison honest; a solar-thermal heating pathway would be a separate sensitivity, not the baseline.

2.3 Uncertainty propagation

Every uncertain input is a triangular `(low, nominal, high)` distribution sourced from the literature (the full parameter table, with a citation on every value, is in `params.py`). A 20,000-trial Monte Carlo propagates these to a 90% interval per route. We propagate stated literature ranges, not assumed Gaussian measurement error. The electrolysis cell voltage and current efficiency of the electrochemical routes are sampled with a physical anti-correlation (a shared operating-severity latent: higher current density raises voltage and lowers efficiency together), avoiding unphysical parameter corners.

2.4 Ranking by paired Monte Carlo

Reading a ranking from five marginal error bars is a statistical error, because the routes share parameters (regolith specific heat, heat recuperation, electrolysis efficiency, liquefaction) that must take the same value in any given world. We therefore run a paired Monte Carlo: one shared parameter draw per trial across all routes, evaluated trial by trial, so we can report P(route A cheaper than route B) and P(route is cheapest / worst).

2.5 Continuous reactor heat loss

A reactor held continuously at 800–1900 C radiates and conducts heat to its surroundings independently of the per-kg sensible heating, a term the sensible-heat-only stages omit. The only measured carbothermal datum (NASA's CaRD brassboard) implies ∼63–93 kWh-thermal per kg O_2 for the reduction step, dominated by this loss at demonstrated (tiny) scale. A scaled, well-insulated plant amortizes it over far more throughput, so the true value is highly scale-dependent. We therefore add a single reactor-loss term (log-uniform, nominal 8, range 2–30 kWh/kg O_2: 2 for a large well-insulated plant, ∼30 approaching the brassboard upper bound) to the four high-temperature routes. The low-temperature water route (sublimation at ∼273 K) is exempt. Hydrogen reduction is also exempt *in modeling* because it is calibrated to Leger 2025, whose full-chain figure already reflects realistic reduction energy; charging it a separate loss term would double-count and break that validation. This term, omitted in an earlier version of the model, is what previously made carbothermal appear cheapest.

3 Validation

3.1 Hydrogen reduction against Leger 2025 (and a second anchor)

Hydrogen reduction is the only route with a clean published electrical figure: 24.3 ± 5.8 kWh/kg LOX, with the reduction step ∼55% and water electrolysis ∼38% of the total (Leger et al., 2025). We built our estimate bottom-up from first principles and independent parameters; the largest tunable term (water-electrolysis efficiency) is set from an independent SOEC source (Hauch et al., 2020; International Energy Agency, 2019), not from Leger's implied value.

- Nominal point estimate: **24.3 kWh/kg LOX**, matching Leger's central value and inside his 1-sigma interval [18.5, 30.1].
- The Monte-Carlo distribution is centered lower (median ∼21, mean ∼22): Leger's value sits near our 85th percentile, not our center. We therefore report agreement as substantial **interval overlap** between two independent estimates, not as a point match; the nominal coincidence should not be over-read.
- Stage shares match: heating + reaction ∼65% (Leger ∼55%); water electrolysis ∼30% (Leger ∼38%).
- A second cross-check: Taylor and Carrier (1993) place the route at ∼26 kWh/kg LOX (cross-technology range 18–35).
- Because the model omits several one-signed terms (Section 9) that would raise totals, the nominal agreement with Leger and the lower distribution center are both consistent with the two estimates sharing a partial system boundary; we do not claim the absolute level is converged.

3.2 MRE against Carr 1963 and terrestrial molten-oxide electrolysis

MRE began as a pure first-principles estimate (nominal 26.5, 90% CI [20.1, 42.0], now including the reactor-loss term). Because it has no published electrical figure, we cross-check it against three sources that unavoidably use *different system boundaries*, so the agreement is weak and bounding rather than a clean anchor: Carr (1963), as reported secondarily by Schreiner et al. (2016), gives ∼26.4 kWh/kg O_2 for lunar MRE; terrestrial molten-oxide electrolysis of iron runs ∼3.7–4.0 MWh/t metal, i.e. ∼9 kWh/kg O_2, a well-insulated lower bound for a different chemistry (Allanore, 2015); and NASA reactor-sizing models place *whole-system* figures (including Joule heating and duty cycle, which our standalone-reactor boundary excludes) at ∼50–120 kWh/kg O_2 (Schreiner et al., 2016). Our estimate falls between these, which given their ∼9–120 spread is corroboration only in a loose sense. We did, however, correct the oxide-melt current efficiency to match *measured* regolith-surrogate data (70–90%, Ir anodes Allanore, 2015) rather than an optimistic assumption, a change that lowered MRE's estimate against the prior narrative.

These are the project's falsifiable tests (`tests/test_validation.py`). Only hydrogen reduction has a like-for-like published electrical figure; the others are first-principles estimates reported with wide intervals and as probabilities, not point claims.

4 Results

Table 1: Electrical-equivalent energy per route (20,000 trials).

Route	Yields	kWh/kg O_2 (nominal)	90% CI	kWh/kg propellant
PSR water mining	LOX+LH_2	14.0	11.8–16.4	**12.5**
Carbothermal (CH_4)	LOX	21.1	16.1–33.1	21.1
Molten-salt (FFC)	LOX	23.3	17.9–36.2	23.3
H_2 reduction (ilmenite)	LOX	24.3	16.4–27.1	24.3
Molten regolith electrolysis	LOX	26.5	20.1–42.0	26.5

The single value is the deterministic nominal (all parameters at their cited value); for the right-skewed routes the Monte-Carlo median can differ by a few kWh/kg, so for plant sizing prefer the median and the interval. The four high-temperature routes carry the reactor heat-loss term (Section 2.5) and so have wide, strongly overlapping intervals; they are not cleanly separable from one another. The water route stands apart below them.

Table 2: Paired-Monte-Carlo dominance.

Route	P(cheapest)	P(worst)
PSR water mining	0.99	0.00
Carbothermal (CH_4)	0.01	0.00
H_2 reduction (ilmenite)	0.01	0.09
Molten-salt (FFC)	0.00	0.15
Molten regolith electrolysis	0.00	0.76

The water route is the cheapest in 99% of paired trials and beats every high-temperature route in >95%. Among the high-temperature routes MRE is the most likely worst (0.76), driven by its reactor loss and cell voltage; the rest are not cleanly separable. The water route is essentially never the worst.

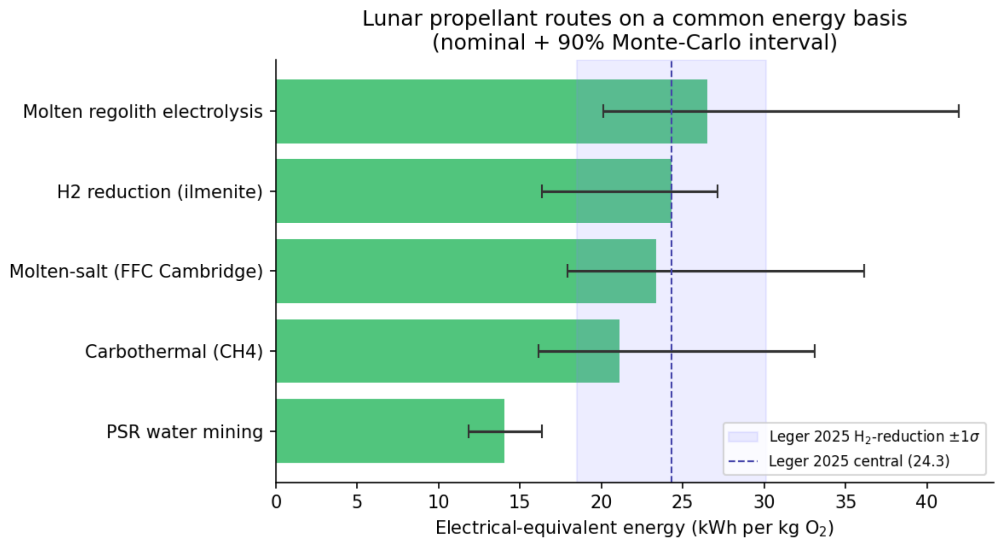

Figure 1: The five routes on a common electrical-energy basis (nominal + 90% Monte-Carlo interval); the shaded band is Leger 2025's 1-sigma for hydrogen reduction.

5 Sensitivity analysis

A one-at-a-time tornado analysis (each parameter swept low-to-high, all others nominal; `python -m lpem -sensitivity <route>`) identifies the dominant uncertainty for each route and, with it, the single highest-value measurement:

Table 3: Dominant sensitivity driver per route.

Route	Dominant driver (swing, kWh/kg O_2)	Next
PSR water	**LH_2 liquefaction (6.0)**	electrolysis eff (2.4), ice grade (1.0)
Carbothermal	**reactor heat loss (28.0)**	reaction enthalpy (3.5), electrolysis eff (2.4)
Molten-salt	**reactor heat loss (28.0)**	current efficiency (5.6), cell voltage (4.5)
MRE	**reactor heat loss (28.0)**	cell voltage (15.9), yield (3.8)
H_2 reduction	O_2 yield (13.3)	heat recuperation (9.1), cp (5.3)

Two practical conclusions. First, the reactor heat-loss term dominates the uncertainty of all three high-temperature routes that carry it; reducing it (better insulation, larger throughput to amortize standing loss) is the single highest-value engineering lever, and measuring it at relevant scale is the highest-value measurement this model identifies (the only datum today is the loss-dominated CaRD brassboard). Second, the water route's verdict hinges on small-scale LH_2 liquefaction, a comparatively well-characterized term. The *ranking* is robust: the water route's lead over every high-temperature route survives across the parameter ranges, precisely because its advantage (no high-temperature reactor) is structural rather than parametric.

6 Findings

Finding 1: The low-temperature polar water route is the most energy-efficient AND the only full-propellant route. It is the cheapest in 99% of paired trials (14.0 kWh/kg O_2, 12.5 per kg of propellant) and is essentially never the worst. The mechanism is structural: water mining operates at the ~273 K sublimation point, so it is the only route that does not run a hot reactor and therefore the only one that escapes the continuous reactor heat-loss penalty

(Section 2.5) that burdens every thermochemical and electrochemical route. It is also the only route that co-produces usable hydrogen, yielding a complete LOX+LH_2 propellant rather than LOX with fuel shipped from Earth. The water route therefore wins on both axes at once. The caveat is honest and route-specific: its dominant cost is small-scale LH_2 liquefaction and (out of this boundary) months-long zero-boil-off storage at 20 K, which could erode but not plausibly erase a >40% energy margin.

Finding 2: Every high-temperature route carries a large, scale-uncertain reactor heat-loss penalty; this is what was missing when carbothermal earlier appeared cheapest. Carbothermal (21.1), molten-salt (23.3), hydrogen reduction (24.3), and MRE (26.5) cluster together with wide, strongly overlapping intervals and are not cleanly separable; MRE is the most likely worst (P 0.76), driven by its reactor loss and full decomposition voltage (~3.5 V, not the ~1.7 V thermodynamic floor nor the 16–34 V Joule-heating voltages quoted in concept papers). An earlier version of this model omitted continuous reactor heat loss and consequently found carbothermal "cheapest"; the only measured carbothermal datum (NASA CaRD, ~63–93 kWh-thermal/kg O_2 for the reduction step, loss-dominated at demonstrated scale) shows that omission was decisive. We charge a de-rated, wide-range loss term to the four high-temperature routes; hydrogen reduction is the exception in modeling only because it is anchored to Leger 2025, whose full-chain figure already reflects realistic reduction energy (charging it again would double-count and break that validation).

Finding 3: The honest separation is low-temperature vs high-temperature, not chemistry. The dominant energy distinction across all five routes is not the specific extraction chemistry but the operating temperature: one low-temperature route (water, ~14) sits far below a cluster of four high-temperature routes (~21–27) whose differences are smaller than their shared reactor-loss uncertainty. For an architect, that reframes route selection: the first-order energy lever is avoiding a high-temperature reactor (or driving down its standing loss), not picking among reduction chemistries.

7 From energy to power plant and landed mass

Energy matters through what it costs to supply. Converting kWh/kg into continuous surface power and the landed mass of the fission-surface-power (FSP) system it implies (`lpem.arch`), even a modest 50 t O_2/yr plant requires ~90–170 kWe, one to ~1.7 units of NASA's planned 100-kWe FY2030 reactor (the cited 40-kWe concept (Oleson et al., 2022) scaled up). Route choice swings landed power-system mass by ~18 t for the same oxygen output: the low-temperature water route needs ~20 t of FSP, while the high-temperature routes need ~30–38 t (carbothermal ~30, MRE ~38), again because of their reactor-loss burden. The direction is robust; the magnitude scales with the FSP specific mass and assumes an all-fission architecture (a solar-plus-storage plant would be dominated by night-survival storage mass, not modeled here).

8 Companion analyses (compute waste-heat integration)

Two companion analyses, kept modular, examine reusing co-located compute waste heat:

- **Low-grade thermal offset** (`WASTE-HEAT-OFFSET.md`). By the Second Law, compute waste heat (~330 K) can supply only low-grade ISRU demand. At the nominal 350 K reject temperature, comfortably above the ~273 K sublimation target, it can in principle cover the water route's full thermal-mining term (~2.1 kWh/kg O_2, ~14% of the route; Figure 2); this offset falls off sharply and excludes the sublimation enthalpy if compute rejects below ~273 K. It cannot supply high-grade reduction/electrolysis heat at all.
- **PSR co-location** (`PSR-COLOCATION.md`). A permanently shadowed region is both a cryogenic radiative sink for compute and the location of the water resource. Under an explicit

radiator energy balance, PSR siting saves (over feasible cases) a median ∼29 t of radiator mass per MW of compute (IQR ∼18–49; Figure 3); in ∼30% of sampled conditions a sunlit 330 K radiator cannot reject at all without active cooling, an extreme form of the same advantage that we report as a feasibility fraction rather than fold into a point estimate. The heat cascade itself is a modest bonus (saves ∼2.7 t reactor for the reference plant; the break-even enabling probability is ∼38% at nominal but ∼47% when propagated, with P(worthwhile once co-located) ∼85%); co-location is justified by the compute siting economics, not the cascade. This is, fittingly, most useful at the low-temperature water route that Finding 1 already favors.

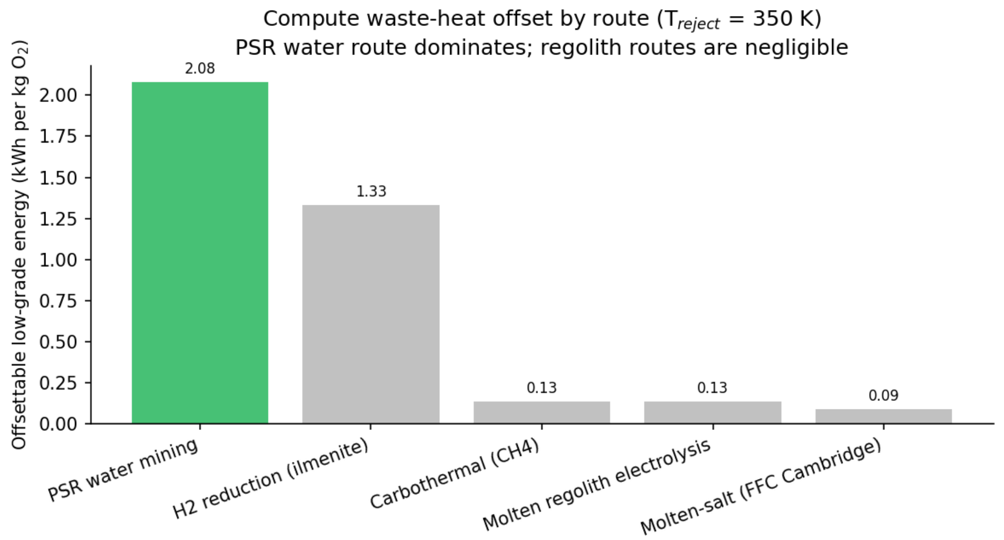

Figure 2: Low-grade, waste-heat-offsettable energy per route ($T_{\mathrm{reject}} = 350$ K).

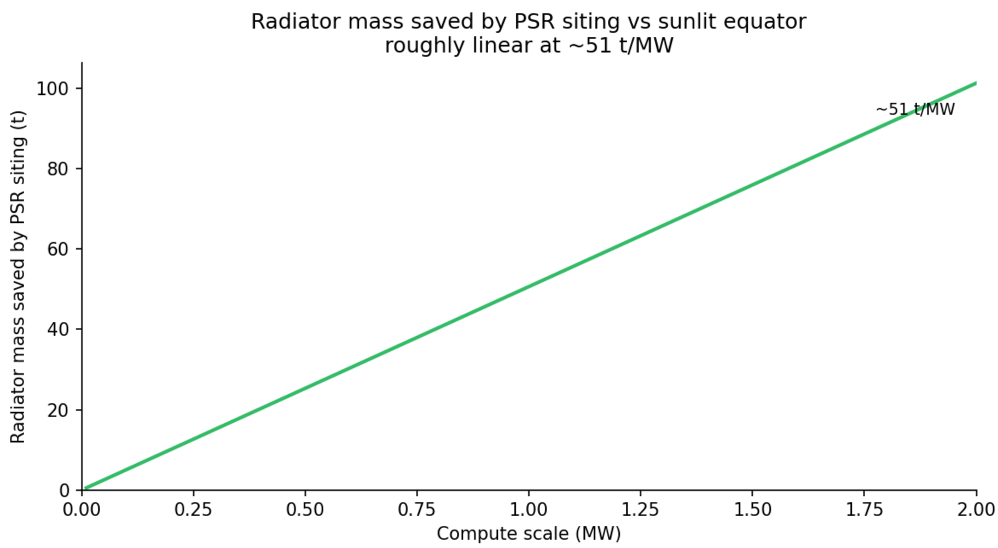

Figure 3: Radiator mass saved by PSR siting vs a sunlit site at nominal parameters (slope ∼51 t/MW); the Monte-Carlo feasible-case median is lower (∼29 t/MW) because the saving is right-skewed and ∼30% of sampled sunlit designs cannot reject at all.

9 Limitations

- Steady-state production energy, power, and FSP landed mass only; capital cost, mobility, comms, site logistics, and demand are out of scope.
- Omitted energy terms are one-signed (they raise totals): standing radiative loss from continuously hot reactors (worst for the hottest route, MRE), comminution/beneficiation, electrolyzer heat-rejection parasitics, and, for the water route, months-long LH_2 zero-boil-off and power delivery into permanent shadow. Absolute kWh/kg figures are best read as lower bounds; the ranking is more robust than the levels.
- Three of five routes lack a direct independent electrical anchor.
- O_2 yield and reaction temperature are sampled independently (a residual idealization).
- Site geography and solar-thermal process heat are not modeled.

10 Reproducibility

```
pip install -e .
python -m lpem                  # Table 1
python -m lpem --dominance      # Table 2
python -m lpem --sensitivity mre   # Section 5 tornado
python -m lpem --plant-tonnes 50   # Section 7
python -m lpem --waste-heat --benefit   # Section 8
python scripts/make_figures.py     # Figures 1-3
pytest                          # 48 tests, incl. the validation anchors
```

References

The full bibliography (with DOIs/URLs) is in `paper/REFERENCES.md` and `paper/references.bib`. Key sources: Leger et al. (2025) (PNAS); Taylor and Carrier (1993); Carr (1963); Allanore (2015) (J. Electrochem. Soc.); Schreiner et al. (2016) (Adv. Space Res.); Kornuta et al. (2019) (REACH); Hauch et al. (2020) (Science); Colaprete et al. (2010) (Science).

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